

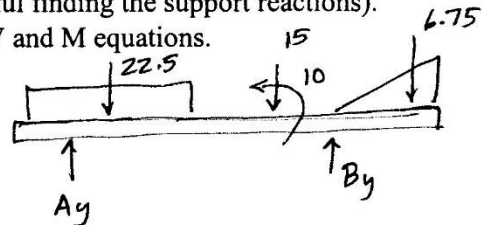
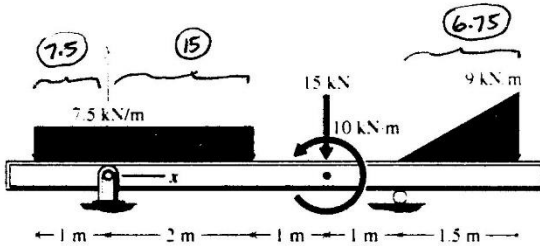
Statics Internal Forces: Shear and Bending Moment Diagrams
(Example Problem 1: Graphical Method plus Write V and M Equations for One Interval)

BE 50 D
 QUIZ: V & M DIAGRAMS
 OCTOBER 26, 1998

NAME: KEY

For the beam loaded as shown below, do the following:

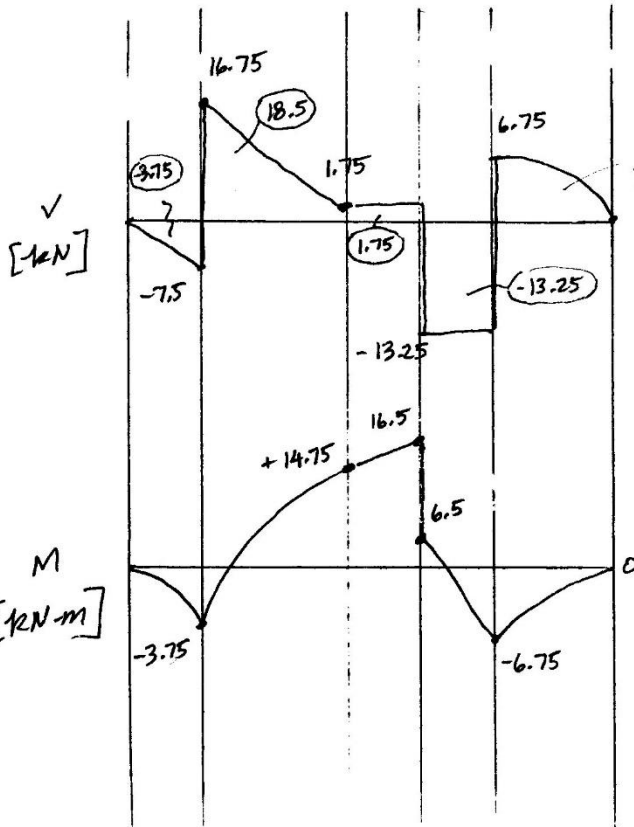
- Draw shear and bending moment diagrams (be careful finding the support reactions).
- For the interval $(1 \leq x \leq 3)$, draw a FBD and write V and M equations.



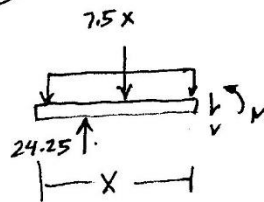
$$+\circlearrowleft \sum M_A = 0; \quad 22.5(1.5) + 15(3) - 10 + 6.75(5) - B_y \cdot 4 = 0$$

$$B_y = 20 \text{ kN}$$

$$A_y = 24.25 \text{ kN}$$



$$\frac{2}{3}(1.5)(6.75) = 6.75$$



$$+\uparrow \sum F_y = 0; \quad 24.25 - V - 7.5x = 0$$

$$V = 24.25 - 7.5x$$

$$+\circlearrowleft \sum M_{cut} = 0; \quad M + 7.5x\left(\frac{x}{2}\right) - 24.25(x-1) = 0$$

$$M = -3.75x^2 + 24.25(x-1)$$

$$M = -24.25 + 24.25x - 3.75x^2$$